

Prefix window 16, queue weight 3, prefix weight 2

The completions load is in [eval.md](#). The locked document is [llminferenceservice.yaml](#).

OpenShift AI **3.4.4**. Four Qwen3-0.6B replicas. Queue weight **3**, prefix weight **2**. vLLM left at `--block-size=32`. The model pods were not restarted.

`blockSizeTokens: 16` is a 64-character window (16×4). The product weights are queue 2 and prefix 3. This run swaps them. A full prefix match still adds to the queue score of the pod that already holds the prefix, so a higher queue weight does not by itself send the next request to an idle pod while that queue is empty.

Client p50 on the first pass stayed between 0.19 s and 0.22 s. Qwen3-0.6B answers faster than a queue forms at these depths.

Pod order is `clwtr, jnqwp, m4ktj, qgbvj`.

Window locked at 16

Scheduler `qwen-l4-kserve-router-scheduler-68c959d4c9-6zwwk, 2026-10-05T15:36:08Z`:
`blockSizeTokens: 16, autoTune: false, queue weight 3, prefix weight 2`.

Family	Shared text	Load	Requests per pod	Hits	Result
linen	59 characters, 10 tokens	24 requests, concurrency 4	6, 7, 3, 8	0	Spread. Under 64 characters.
pearl	95 characters, 16 tokens	24, concurrency 4	0, 0, 24, 0	0	Pinned. One picker block, no KV block.
topaz	167 characters, 55 tokens	24, concurrency 4	0, 0, 24, 0	736	Pinned. $736 = 23 \times 32$.
crimson	765 characters, 96 tokens	24, concurrency 4	0, 0, 0, 24	2208	Pinned. $2208 = 23 \times 96$.
indigo	96 characters, 27 tokens	96, concurrency 32	0, 96, 0, 0	0	Pinned. One picker block, no KV block.
maroon	341 characters, 97 tokens	96, concurrency 32	0, 0, 0, 96	9120	Pinned. $9120 = 95 \times 96$.

`pearl` is the gap. Ninety-five characters complete one 64-character picker block and stay at 16 Qwen tokens, under one vLLM block of 32. All 24 requests landed on `m4ktj` and `vllm:prefix_cache_hits_total` stayed 0. `indigo` is the same shape at 32 concurrent users: all 96 requests on `jnqwp`, hit counter 0.

`linen` at 59 characters does not complete a picker block and used all four pods.

Field omitted

The first pass removed the prefix-cache-scorer parameters, which is the 3.4 default. Scheduler qwen-l4-kserve-router-scheduler-599b8f7c4d-zmjwh at 2026-10-05T15:29:02Z logged BlockSize is not positive, using default value default=16, then autoTune: true and blockSizeTokens: 16. The same log did not print a later block size. The requests followed a 128-character window, which is 32×4 , the vLLM block size with autotune left on.

Family	Shared text	Load	Requests per pod	Hits	Result
Unique prompts	none	24, concurrency 4	4, 5, 9, 6	0	Spread
marble	61 characters, 9 tokens	24, concurrency 4	3, 7, 4, 10	0	Spread
amber	95 characters, 16 tokens	24, concurrency 4	7, 5, 7, 5	0	Spread
jasper	96 characters, 27 tokens	24, concurrency 4	6, 4, 6, 8	0	Spread
onyx	105 characters, 41 tokens	24, concurrency 4	7, 4, 3, 10	192, 96, 64, 288	Spread, hits on every pod
velvet	96 characters, 14 tokens	96, concurrency 32	18, 24, 27, 27	0	Spread
Unique, busy	none	96, concurrency 32	21, 25, 27, 23	0	Spread
cobalt	159 characters, 45 tokens	24, concurrency 4	0, 24, 0, 0	736	Pinned. $736 = 23 \times 32$
nickel	229 characters, 33 tokens	24, concurrency 4	0, 0, 0, 24	736	Pinned. $736 = 23 \times 32$
copper	278 characters, 40 tokens	96, concurrency 32	96, 0, 0, 0	3040	Pinned. $3040 = 95 \times 32$
granite	317 characters, 40 tokens	24, concurrency 4	24, 0, 0, 0	736	Pinned. $736 = 23 \times 32$
basalt	341 characters, 97 tokens	24, concurrency 4	0, 0, 24, 0	2208	Pinned. $2208 = 23 \times 96$
saffron	261 characters, 97 tokens	96, concurrency 32	96, 0, 0, 0	9120	Pinned. $9120 = 95 \times 96$

onyx is under 128 characters and over 32 tokens. The picker did not keep the family together. Each pod that saw the prompt more than once stored one KV block: the hit columns are (requests - 1) $\times 32$.

amber at 95 characters spread on this pass. The same length, pearl, pinned after autoTune was turned off and blockSizeTokens was set to 16.

What the swap changed

A prefix the picker scores still stays on one pod when the queues are empty. Queue weight 3 and prefix weight 2 did not move pearl, indigo, topaz, crimson, or maroon. Concurrency 32 did not move indigo or maroon.

The swap shows up when the picker scores no block. With the field omitted and autotune on, that line was 128 characters: amber, jasper, and velvet used all four pods, and cobalt at 159 characters pinned. With the window locked at 16, that line is 64 characters: linen spread, and pearl at 95 characters pinned with a hit counter of 0.